Siddharth Chandak | CV

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Education

- **Stanford University** — **Stanford, CA, USA**
  Ph.D., Electrical Engineering — *2021–present (expected Sep. 2026)*
  Advisor: Prof. Nicholas Bambos
- **Stanford University** — **Stanford, CA, USA**
  M.S., Electrical Engineering, GPA: 4.15/4 — *2021–2023*
- **Indian Institute of Technology Bombay** — **Mumbai, India**
  B.Tech., Electrical Engineering, CPI 9.89/10 — *2017–2021*
  Project Advisor: Prof. Vivek S. Borkar
  - President of India Gold Medal
  - Honours in Electrical Engineering & Minor in Computer Science

Research Interests

Multi-Agent Learning and Control: My research focuses on multi-agent systems and games, with an emphasis on developing algorithms for distributed learning of equilibria for unknown systems. I also study problems in game control, where the aim is to steer players toward equilibria that satisfy both the local and global objectives.

Stochastic Approximation: I am interested in developing new frameworks for stochastic approximation schemes which enable the design and analysis of algorithms in optimization and reinforcement learning. In addition, I work on obtaining tighter finite-time guarantees for reinforcement learning algorithms.

Selected Awards

- Centennial Teaching Assistant Award from Stanford University in 2025.
- Stanford Graduate Fellowship from 2021 to 2026.
- President of India Gold Medal at IIT Bombay for the highest GPA among graduating students in 2021.
- Prof. K. C. Mukherjee Award at IIT Bombay for best B.Tech. project among EE students in 2021.
- Among the top 50 candidates in the Indian National Olympiads in Chemistry and Physics and chosen to attend respective Selection Camps for International Olympiads in 2017.

Publications

Submitted Articles

1. **S. Chandak**, "Concentration and Mean-Square Bounds for Contractive Stochastic Approximation: A Unified Elementary Approach", *Submitted to Mathematics of Control, Signals, and Systems*.
2. A. Kar, **S. Chandak**, R. Singh, S. Sinhahajari, E. Moulines, S. Bhatnagar, and N. Bambos, "Policy Gradient Methods for Non-Markovian Reinforcement Learning", *Submitted to NeurIPS*, 2026.
3. R. Singh, **S. Chandak**, V. S. Borkar, E. Moulines, and N. Bambos, "Regret and Sample Complexity of Online Q-Learning via Concentration of Stochastic Approximation with Time-Inhomogeneous Markov Chains", *Submitted to NeurIPS*, 2026.
4. **S. Chandak**, A. Yadav, A. Ozgur, and N. Bambos, "Heavy-Tailed and Long-Range Dependent Noise in Stochastic Approximation: A Finite-Time Analysis", *Submitted to IEEE Transactions on Automatic Control*.

5. A. Kar, **S. Chandak**, R. Singh, E. Moulines, S. Bhatnagar, and N. Bambos, "High-Probability Bounds for SGD under the Polyak-Łojasiewicz Condition with Markovian Noise", *Submitted to SIAM Journal on Optimization*.

Journal

1. **S. Chandak**, "$O(1/k)$ Finite-Time Bound for Non-Linear Two-Time-Scale Stochastic Approximation", *IEEE Transactions on Automatic Control*, 2026.
2. **S. Chandak**, "Non-Expansive Mappings in Two-Time-Scale Stochastic Approximation: Finite Time Analysis", *SIAM Journal on Control and Optimization*, to appear.
3. **S. Chandak**, I. Bistritz and N. Bambos, "Choose Your Battles: Distributed Learning Over Multiple Tug of War Games", *IEEE Transactions on Automatic Control*, 2026.
4. **S. Chandak**, I. Bistritz and N. Bambos, "Learning to Control Unknown Strongly Monotone Games", *IEEE Transactions on Control of Network Systems*, 2026.
5. **S. Chandak** and V. S. Borkar, "A Concentration Bound for TD(0) with Function Approximation", *Stochastic Systems*, 2026.
6. **S. Chandak**, P. Shah, V. S. Borkar and P. Dodhia, "Reinforcement Learning in Non-Markovian Environments", *Systems and Control Letters*, 2024.
7. **S. Chandak**, V. S. Borkar and H. Dolhare, "A Concentration Bound for LSPE(λ)", *Systems and Control Letters*, 2023.
8. **S. Chandak**, V. S. Borkar and P. Dodhia, "Concentration of Contractive Stochastic Approximation and Reinforcement Learning", *Stochastic Systems*, 2022.
9. S. U. Haque, **S. Chandak**, F. Chiariotti, D. Günduz and P. Popovski, "Learning to Speak on Behalf of a Group: Medium Access Control for Sending a Shared Message," *IEEE Communications Letters*, 2022.
10. V. S. Borkar and **S. Chandak**, "Prospect-theoretic Q-learning", *Systems and Control Letters*, 2021.
11. **S. Chandak**, F. Chiariotti and P. Popovski, "Hidden Markov Model-Based Encoding for Time-Correlated IoT Sources", *IEEE Communications Letters*, 2021.

Conference

1. **S. Chandak**, R. Tamizholi, and N. Bambos, "Last-Iterate Guarantees for Learning in Co-coercive Games", *To be presented at IEEE Conference on Decision and Control (CDC)*, 2026.
2. **S. Chandak**, S. U. Haque and N. Bambos, "Finite-Time Bounds for Two-Time-Scale Stochastic Approximation with Arbitrary Norm Contractions and Markovian Noise", *IEEE Conference on Decision and Control (CDC)*, 2025.
3. **S. Chandak**, I. Thapa, N. Bambos and D. Scheinker, "Optimal Control for Remote Patient Monitoring with Multidimensional Health States", *IEEE International Conference on Communications (ICC)*, 2025.
4. **S. Chandak**, I. Thapa, N. Bambos and D. Scheinker, "Tiered Service Architecture for Remote Patient Monitoring", *IEEE Healthcom*, 2024.
5. **S. Chandak**, I. Bistritz and N. Bambos, "Tug of Peace: Distributed Learning for Quality of Service Guarantees", *IEEE Conference on Decision and Control (CDC)*, 2023.
6. **S. Chandak**, I. Bistritz and N. Bambos, "Equilibrium Bandits: Learning Optimal Equilibria of Unknown Dynamics", *International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*, 2023.

Invited Talks

- Learning to Control Unknown Strongly Monotone Games
  - Seminar at Automatic Control Lab, EPFL, March 2026
- O(1/k) Finite-Time Bound for Non-Linear Two-Time-Scale Stochastic Approximation

- INFORMS Annual Meeting, October 2025.

- Learning to Control Unknown Multi-Agent Systems
  - Department of Electrical Engineering Seminar, IIT Bombay, September 2025.
  - STCS Seminar, Tata Institute of Fundamental Research (TIFR) Mumbai, September 2025.

- Non-Expansive Mappings in Two-Time-Scale Stochastic Approximation
  - INFORMS Applied Probability Society Conference, July 2025.

Teaching Experience

Course Instructor at Stanford University

- ENGR 76: Information Science and Engineering *Winter 2026*
  - Course aimed at introducing undergraduate students at Stanford to the fundamentals behind modern information and communication systems.

Teaching Assistant at Stanford University

- ENGR 76: Information Science and Engineering *Spring 2024, 2025*
  Instructor: Prof. Ayfer Ozgur
  - Head TA, leading a team of 10+ TAs for a class of around 300 students.
  - Helped restructure the class by redesigning projects, and designing exams and discussion sessions.
  - Awarded the Centennial Teaching Assistant Award for contributions to the course.
- MS&E 130: Information Networks and Services *Winter 2023, 2025, 2026*
  - Delivered two guest lectures on queueing and scheduling.
- EE 263: Linear Dynamical Systems *Autumn 2024*
- MS&E 232: Introduction to Game Theory *Winter 2024*

Teaching Assistant at IIT Bombay

- MA106: Linear Algebra *Spring 2020*
- MA105: Calculus *Fall 2019*
- PH108: Basics of Electricity and Magnetism *Spring 2019*
- PH107: Quantum Physics and Applications *Fall 2018*

Experience

- **Uber** | *Risk and Trusted Identity* Jun. - Sep. 2025
  PhD Machine Learning Engineering Intern
  Developed machine learning models for fraud detection.
- **Morgan Stanley** | *Institutional Equities Division* Jun. - Aug. 2024
  Quantitative Finance Summer Associate
  Developed a statistical factor model for the Central Risk Book desk in Institutional Equities Division.
- **Aalborg University** | *Supervisor - Prof. Petar Popovski* May - Jul. 2021
  Summer Intern
  Proposed a multi-agent multi-armed bandit based approach for transmission of a shared message over multiple shared channels while avoiding collision.
- **Aalborg University** | *Supervisor - Prof. Petar Popovski* Apr. - Jul. 2020
  Summer Intern
  Proposed encoding and decoding scheme for transmitting short IoT packets with time correlation across a noisy channel by modeling source dynamics using Hidden Markov Models.

- **Imperial College London |** *Supervisor - Prof. Nick S. Jones* May - Jul. 2019
  Summer Intern
  Investigated the difference between social networks in UK, ICL and "Hackspace" - a smaller technical community at ICL, by analyzing survey data on friendships, and modeling social networks using Stochastic Block Models.